A Lock Free, Non-blocking, In Line Processing Architecture

for High Throughput and Regulatory Compliant Blockchain Trading Applications

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Motivation

- Regulated DeFi (R-DeFi) needs trading systems that are **simultaneously** high-throughput **and** regulatory compliant[1].
- Off-chain compliance checks (KYC/AML, trade surveillance) and on-chain settlement pull system design in **opposite directions**[2].
- Existing blockchain trading architectures are not empirically benchmarked against production-grade throughput and latency targets[3].

> **Research question:** which architectural pattern maximises throughput while preserving compliance guarantees, under real blockchain constraints (nonce ordering, gas limits, block capacity)?

[1] D. A. Zetzsche, D. W. Arner, R. P. Buckley, "Decentralized Finance," Journal of Financial Regulation, 2020.
[2] Financial Action Task Force, "FATF Guidance on Virtual Assets and Virtual Asset Service Providers," 2024.
[3] M. Bez, G. Fornari, T. Vardanega, "The Scalability Challenge of Ethereum: An Initial Quantitative Analysis," IEEE SOSE, 2019.

Approach

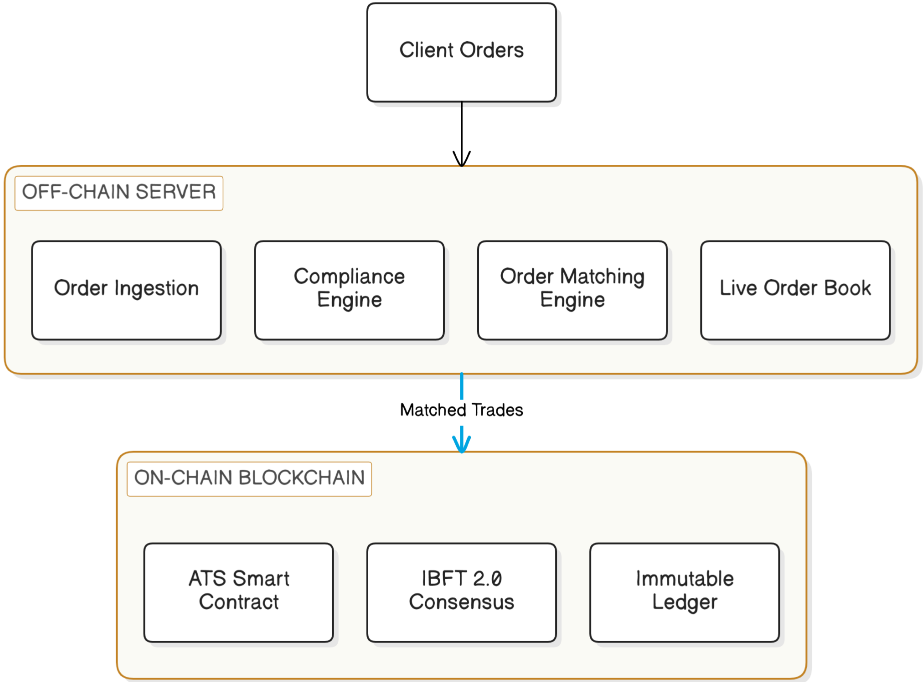
- Two-layer **hybrid architecture**: off-chain compliance / order layer decoupled from on-chain settlement layer.
- **Ablation study** of five progressively optimised architectural variants, deployed and measured on a **live production trading system**.
- Variants isolate the effect of: asynchronous submission, multi-threading, multi-wallet parallelism, and symbol-level sharding with lock-free coordination.
- Empirical measurement of throughput, latency (avg / P99), and block utilisation under identical load.

Two-Layer Hybrid Architecture

Off-chain compliance, matching, and order-book components are decoupled from on-chain settlement — keeping regulatory logic mutable while the ledger stays immutable[1].

Five architectural stages studied:

1. Sequential, blocking, single wallet
2. Asynchronous, single wallet
3. Multi-threaded, asynchronous, single wallet
4. Unsynchronised multi-threaded, multi-wallet
5. **Symbol-sharded, lock-free, multi-wallet (proposed)**



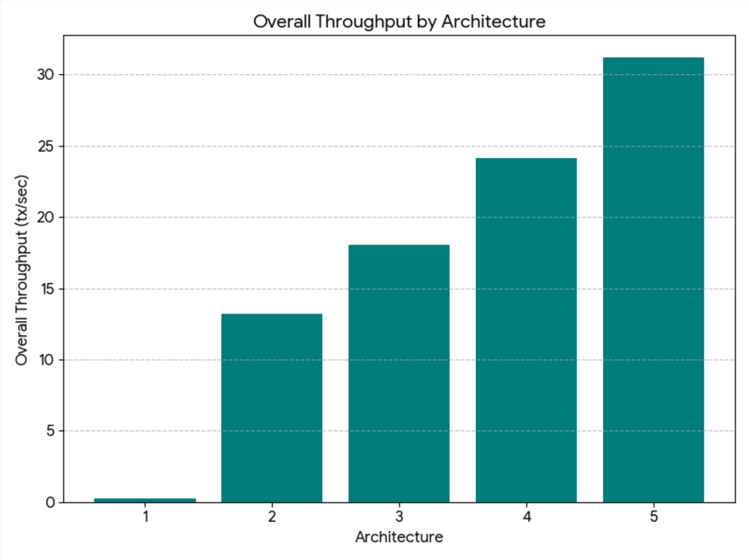
[1] R. Jamithireddy et al., "A Hybrid Blockchain Architecture for Enterprise Adoption," 2025.

Key Findings

- **Nonce mechanism** limits naive parallelism to only a **36% throughput gain** — Ethereum's strict per-account nonce ordering is a hard serialisation point[1].
- Circumventing nonce ordering with multiple wallets **without** careful coordination causes a **9× increase in latency** due to lock contention.
- Bottlenecks **migrate**: solving one on-chain constraint exposes a new server-side concurrency crisis.
- Our **lock-free, symbol-sharded** submission model (inspired by the **LMAX Disruptor** HFT architecture[2]) removes contention entirely by partitioning work per trading symbol.

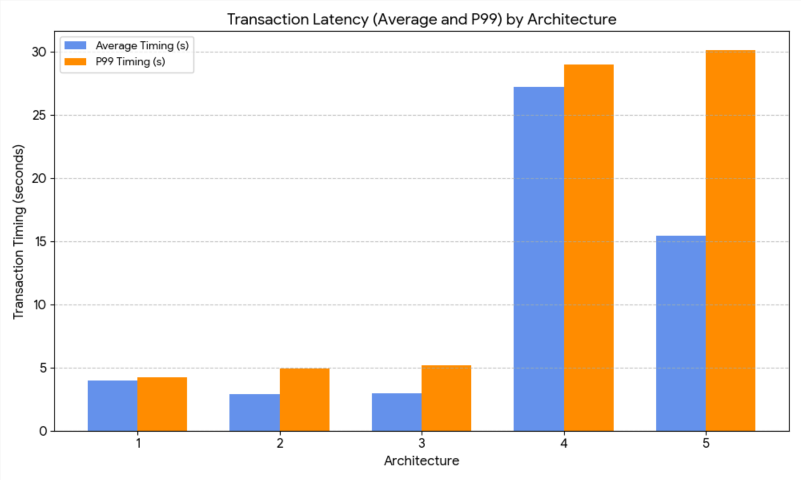
[1] K. Hu, Z. Zhou, X. Li, Z. Ren, "Research on Nonce-Pool-Based Acceleration Scheme for Blockchain Transaction Writing," IEEE Access, 2023.
[2] M. Thompson, D. Farley, M. Barker, P. Gee, A. Stewart, "LMAX Disruptor: High Performance Alternative to Bounded Queues for Exchanging Data Between Concurrent Threads," LMAX Exchange, 2011.

Performance Results

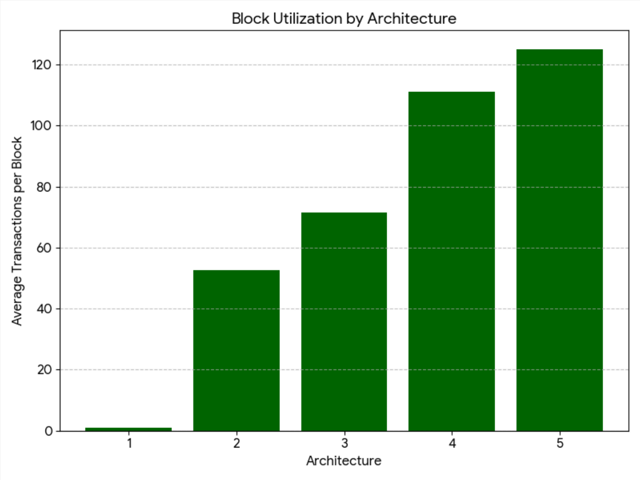


2.8× improvement over conventional design | **125×** gain over naive baseline | **44%** latency reduction at peak throughput

Latency & Block Utilisation



Latency (avg / P99)

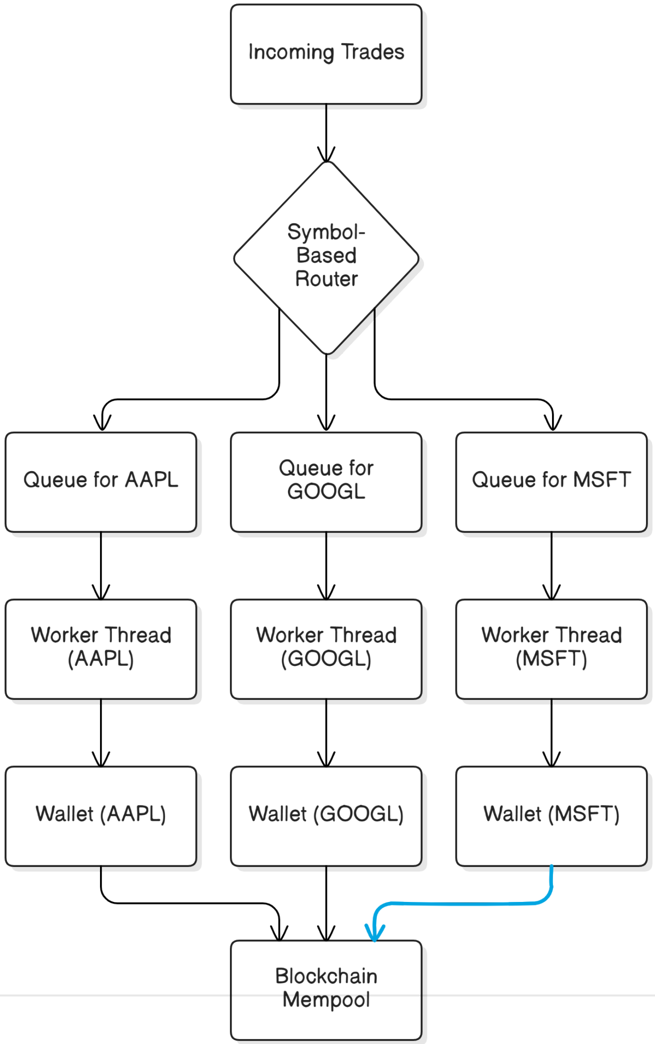


Block utilisation

Symbol-Sharded, Lock-Free Design

Trades are routed by symbol to independent queues and worker threads, each owning a dedicated wallet — removing cross-symbol lock contention entirely, following the LMAX ring-buffer partitioning approach (see Key Findings), and avoiding the cross-shard coordination pitfalls seen in general-purpose blockchain sharding[1].

> Achieves **31.19 tx/sec**, the highest throughput across all five studied variants.



[1] E. Fynn, F. Pedone, "Challenges and Pitfalls of Partitioning Blockchains," IEEE/IFIP DSN-W, 2018.

Production System

- **Upstream** is Horizon Globex's live platform, running **Architecture 5** (symbol-sharded, lock-free) in production.
- Off-chain server handles order ingestion, KYC/AML, and matching.
- Settlement via permissioned **IBFT 2.0** Ethereum and the `ATS.sol` contract.
- Upstream supports **MERJ**, a tier-one regulated stock exchange, alongside US automated trading system deployments.

Conclusions

- A symbol-sharded, lock-free architecture achieves **31.19 tx/sec**, the highest throughput across all five studied variants.
- Naive concurrency fixes (multi-threading, multi-wallet) expose new bottlenecks rather than resolving throughput ceilings.
- Regulatory-compliant, high-throughput blockchain trading is achievable through careful **architectural co-design**, not brute-force parallelism.
- Future work: adaptive sharding and cross-chain generalisation.

Thank You

Questions?

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Code: github.com/HorizonFintex/blockchain-trading-architectures